

Chemistry Paper 26: Detailed Marking Report

Unit 11 Environmental Chemistry Air, sections 11.1 and 11.2 (27 marks) + Revision: Unit 8 Energetics (11 marks)
Seemab · Class 9 · Attempt date 2026-09-30 · Paper: 38 marks, 1h30m, no optional questions

Final Score: 37.5 / 38 · 98.7% · Grade A+

New material (Unit 11): **26.5 / 27** | Revision (Unit 8): **11 / 11** | MCQs: **10 / 10**

This is her best paper of the year, ahead of Physics exam-15 (97.8%). Two markers worked this script independently, the second without sight of the first, and reached the same total of 37.5.

The thing this paper was built to test, and she passed it

Q.6(v) was set for one reason. On Chemistry exam-25 her largest single loss was Q.7, where the reaction was reversed and she reused the worked example's value instead of re-deriving the direction of ΔH . So Q.6 walked her up to it: define enthalpy, write the formula, calculate forward, say whether it is exothermic, and only then asked for the reverse reaction.

She wrote the reverse equation out,

recalculated it from the bond energies from scratch

as $(2 \times 567) - (436 + 155) = 1134 - 591$, and gave

$\Delta H^\circ = +543 \text{ kJ/mol}$

with the sign explicit and the unit present. Then the reason: *"It is positive bcuz the reverse reaction absorbs the same amount of energy that the foward reaction releases."*

She did not flip a sign and hope. She derived it twice and got the same answer both ways. The exam-25 error does not appear anywhere on this script, and every line of Q.6 carries its unit, which was the other exam-25 loss.

Unit 8 revision: 11 / 11.

Section A: MCQs ($10 \times 1 = 10$)

Question	Key	Hers	Note
1 to 10	c a d b c a d b c a	C A D B C A D B C A	10 / 10. Every one correct, none left blank, all ten numbered in her own hand. Both markers ran the shift guard at offsets 0, ± 1 and ± 2 : offset 0 gives 10/10 and every other offset gives 0, so there is no off-by-one misread. This sits exactly on her MCQ baseline.

Section B: Short answers ($3 \times 4 = 12$)

Q	Mark	What she wrote, and how it was marked
2	4 / 4	Both definitions word for word from her book. <i>"The envelope of gases & water vapour surrounding the planet Earth is known as Atmosphere."</i> and <i>"Anything that is in air, water or soil which has a harmful effect on some part of the environment is known pollutant."</i> Nitrogen 78%, oxygen 21%, water vapour 4%, all three right.
3	4 / 4	Six pollutants named, then for sulphur oxides SO_2 , SO_3 with <i>"Power stations and industries using fossil fuels"</i> and <i>"lung cancer, acid rain"</i> ; for nitrogen oxides NO , NO_2 with <i>"Exhaust fumes of motor vehicles"</i> and <i>"Coughs, headaches"</i> . Formulas, sources and two effects each, all correct. She listed carbon dioxide as the sixth where Figure 11.2 has particulates. That quarter mark was given to her. See the note below: the fault is the book's and ours, not hers.
4	3.5 / 4	(i) Correct and complete: produced by incomplete combustion, undetectable because it is <i>"colourless and odorless"</i> . (ii) Correct: <i>"CO combines strongly with hameoglobin to form carbonyhamoglobin. This reduces the blood's ability to carry oxygen around the body."</i> Full marks, see the handwriting note below. (iii) lost ½ mark. She wrote <i>"Ground level ozone is called a secondary pollutant bcuz it is emitted directly."</i> The word not is missing, and both markers checked that line magnified: there is no "not" and no insertion mark. As written, the reason she gives is the opposite of the truth. Her very next sentence is completely correct, <i>"It is formed in the atmosphere when nitrogen onides & hydrocarbons react in sunlight",</i>

which is exactly what her book says a secondary pollutant is, so the mechanism earned ½ and the misstated reason lost ½.

Section C: Long answers (2 × 8 = 16)

Q	Mark	What she wrote, and how it was marked
5	8 / 8	(i) Four rows complete and all correct: carbon monoxide, sulphur dioxide, nitrogen oxide and methane, each with a real source and a real harmful effect from her book. (ii) <i>"Incomplete combustion occurs when a fuel burns with an insufficient supply of O₂. It produces carbon monoxide (CO)."</i> (iii) She explained the closed room fully, then answered the second half too: <i>"Precaution: Keep a window open to provide ventilation."</i> Worth noticing: on Quran exam-03 nine days ago she answered only half of a two-part question. Here the question asked for an explanation <i>and</i> a precaution, and she gave both. That habit is fixed.
6	8 / 8	(i) Definition plus both signs, correct. (ii) Formula correct. (iii) Full working with the formula, the substitution and $\Delta H^\circ = -543 \text{ kJ/mol}$ with its unit. (iv) Exothermic, with the reason. (v) The re-test item, fully correct: see the box at the top. Every line carries its unit.

The two things to carry forward

1. Your letter x reads as an n. This one could cost you real marks.

Across this script you wrote "onide" for oxide, "mononide" for monoxide, "dionide" for dioxide, "enothermic" for exothermic and "carbonyhamoglobin" for carboxyhaemoglobin. Every one of these is the same letter, written the same way.

This is not spelling. Your chemistry is right everywhere, your formulas are right, and you picked carboxyhaemoglobin correctly in MCQ 5, so you know all of these. The problem is the shape: your x is rounded and open, so it reads as n. We could tell what you meant because we compared it with the word "Oxygen" on your own page 3, where the same shape appears in a word that can only be oxygen.

A board examiner will not do that.

They have never seen your handwriting and they mark hundreds of scripts. If they read "enothermic" they may mark it wrong.

Close your x: two strokes that cross clearly.

Nothing else in your writing needs to change.

2. The missing "not" in Q.4(iii)

You meant "because it is

not

emitted directly" and you wrote "because it is emitted directly". One dropped word turned your answer into its opposite. You rescued it with the next sentence, which is why you kept half the mark.

When a question asks *why* something is called something, read your own sentence back once before moving on. This is the same habit that saves the units mark: write it, then look at it once.

Not your mistake: the book contradicts itself about the six air pollutants

Q.3 asked for the six pollutants in Figure 11.2. You answered carbon monoxide, nitrogen oxides, ground level ozone, methane, sulphur oxides and
carbon dioxide

Figure 11.2 on your page 156 shows sulphur oxides, carbon monoxide, nitrogen oxides, methane, ground level ozone and
particulates

, with no carbon dioxide. But the numbered sections on the same pages give a different six: 11.2.1 sulphur oxides, 11.2.2 nitrogen oxides,

11.2.3 carbon dioxide

, 11.2.4 carbon monoxide, 11.2.5 methane, 11.2.6 ground level ozone, and particulates never get a section at all.

So your answer is right by the contents and wrong by the picture, and our own paper made it worse by naming the sections 11.2.1 to 11.2.6 as the syllabus. Our question was defective, so the mark is yours. For the exam, know both:

particulates

belongs in the figure's six,

carbon dioxide

has its own section.

The same page still prints the heading "10.2 AIR POLLUTANTS" where it means 11.2. Write 11.

Marked 30 September 2026. First pass and an independent blind second marking, the second marker working from the scan and the textbook without sight of the first. Both reached 37.5 / 38. The two markers initially differed on where to place one half mark, between Q.4(ii) and Q.4(iii); that was settled by comparing her letterforms on the scan directly, which showed the Q.4(ii) wording to be her handwriting rather than an error. Every answer was re-derived from her own textbook rather than taken from the key. Answer key issued with this report.